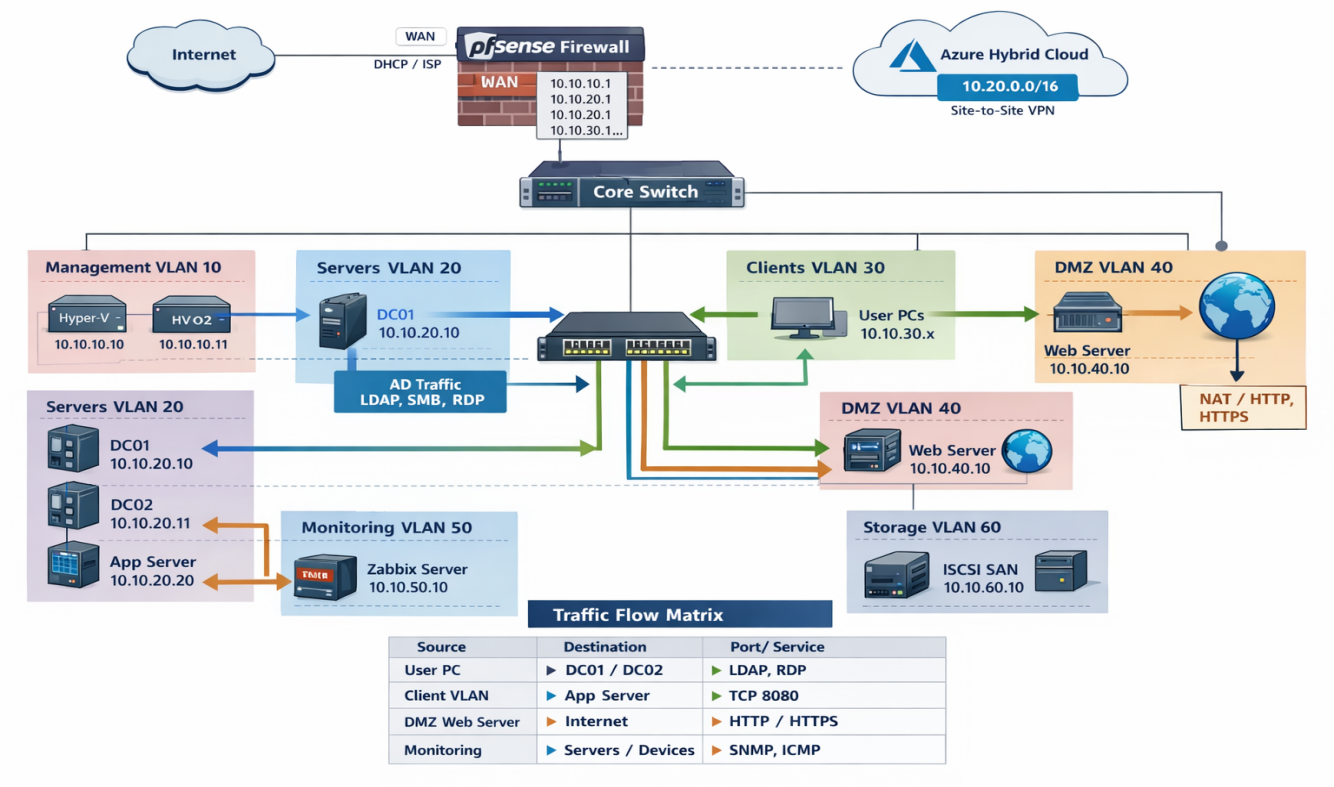
Architect-Level Network Design (Advanced)



1. Packet Flow (Step-by-Step Traversal)

Example: Client accessing Active Directory 1. ■■ Client (10.10.30.100) sends request 2. ■ Packet reaches Gateway (pfSense - 10.10.30.1) 3. ■ Firewall rule check (Allow LDAP/Kerberos) 4. ■ Routed to Server VLAN (10.10.20.0/24) 5. ■ Delivered to DC01 (10.10.20.10) 6. ■ Response follows reverse path Flow: Client → Gateway → Firewall → Routing → Server → Response

2. Traffic Flow Matrix (with Icons)

Source	➡■	Destination	Protocol	Purpose
■■ Client	➡■	■ DC01/DC02	LDAP/Kerberos	Authentication
■■■ Admin	➡■	■■ Hyper-V	RDP	Management
■ Zabbix	➡■	■■ Servers	SNMP/ICMP	Monitoring
■ DMZ Web	➡■	■ Internet	HTTP/HTTPS	Public Access

3. Flow Explanation

All traffic flows are controlled via pfSense firewall. Inter-VLAN communication is denied by default and only allowed based on specific rules. Each packet undergoes: - Source validation - Firewall rule inspection - Routing decision - Destination delivery